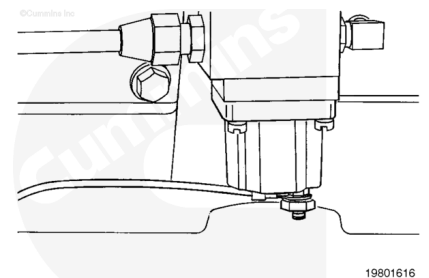


Trainings ▾

General Information

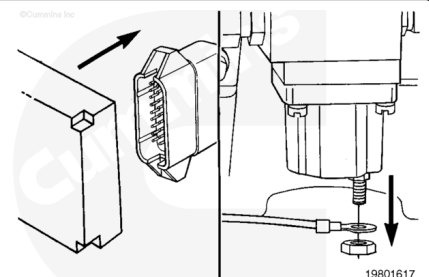
The step timing control (STC) circuit consists of the driver wire connected to pin 20 of the main engine harness connector and an oil control valve that is controlled by a solenoid. The STC driver wire is connected to this solenoid. The oil control valve is normally closed. When the oil control valve is closed, the engine is in normal timing.



Resistance Check

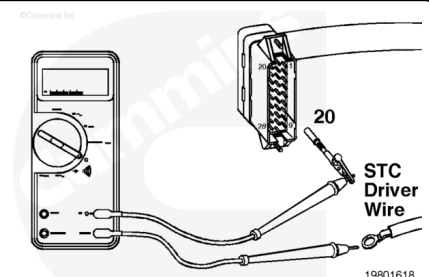
Disconnect the ECM connector.

Disconnect the STC driver wire terminal from the STC solenoid.



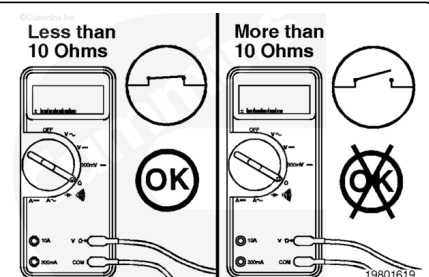
Select the resistance function on the multimeter.

Touch one of the multimeter leads to pin 20 of the main engine harness connector. Touch the other multimeter lead to the ring terminal on the end of the STC driver wire.



Measure the resistance.

The multimeter **must** show less than 10 ohms, which is a closed circuit. If the circuit is **not** closed, then there is an open circuit in the STC driver wire.

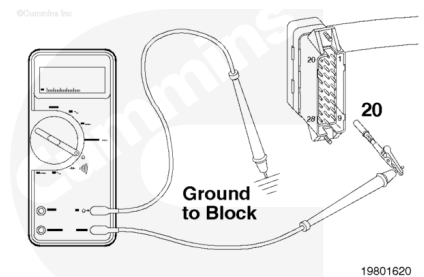


Repair the main engine harness, or, if necessary, replace it.
Refer to Procedure 019-228
(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or
019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-
043.html).

Check for Short Circuit to Ground

Make sure the STC driver wire is disconnected from the solenoid.

Touch one of the multimeter leads to pin 20 of the main engine harness connector. Touch the other multimeter lead to a good, clean surface on the engine block.



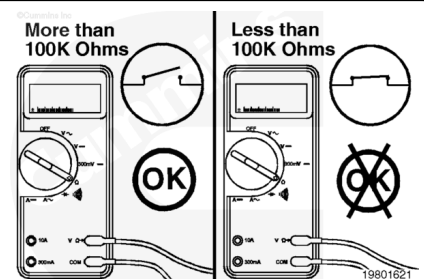
Measure the resistance.

The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short circuit between the wire connected to pin 20 and chassis ground.

Repair the main engine harness, or, if necessary, replace it.

Refer to Procedure 019-228

(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or
019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-
043.html).

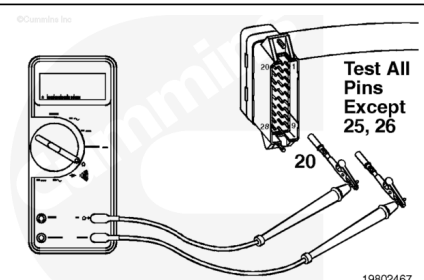


Check for Short Circuit from Pin to Pin

Disconnect the ECM connector and the C5 and C6 connectors.

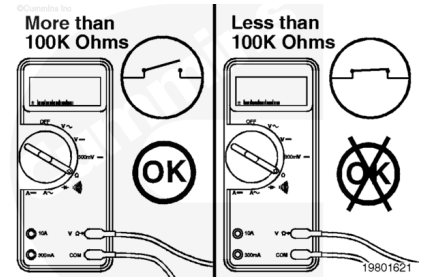
Check for a short circuit between pin 20 of the main engine harness connector and **all** other pins in the connector **except** pins 25 and 26.

Touch one of the multimeter leads to pin 20 of the main engine harness connector. Touch the other multimeter lead to **all** other pins in the connector **except** pins 25 and 26, one at a time.



Measure the resistance.

The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short between the wires connected to pin 20 of the main engine harness connector and **any** other pin that measured less than 100k ohms.



Repair the main engine harness, or, if necessary, replace it.

Refer to Procedure 019-228

(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or
019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-
043.html).